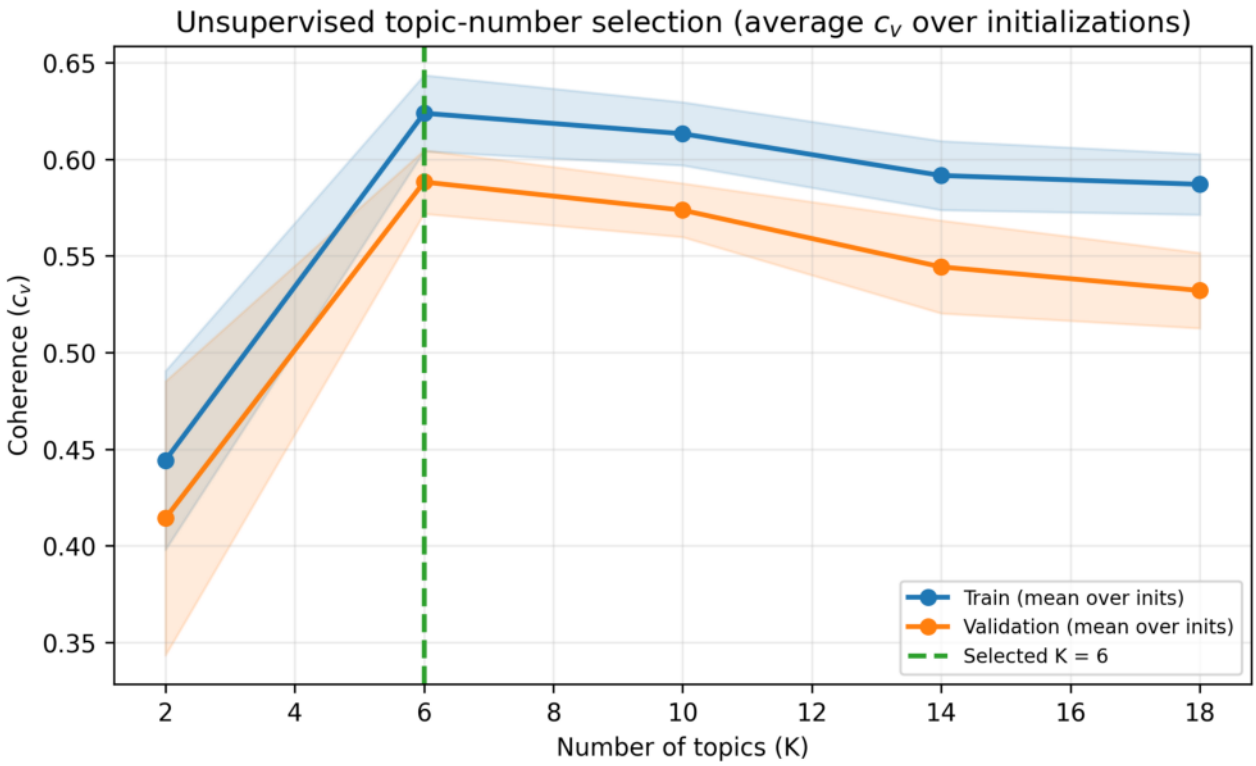
LDA topic-number selection and stability — revision

Revised CNIO/DB1 embeddings · cosine · Hungarian vs LDA_model_vdec25
Same report structure as results/lda/stability/lda_stability_report.*

1. Unsupervised selection of K → K = 6



Candidates K ∈ {2,6,10,14,18}; 5 inits; train/val 70/30; maximise mean validation c_v → K = 6.

2. Level 1 — Stability to random seed

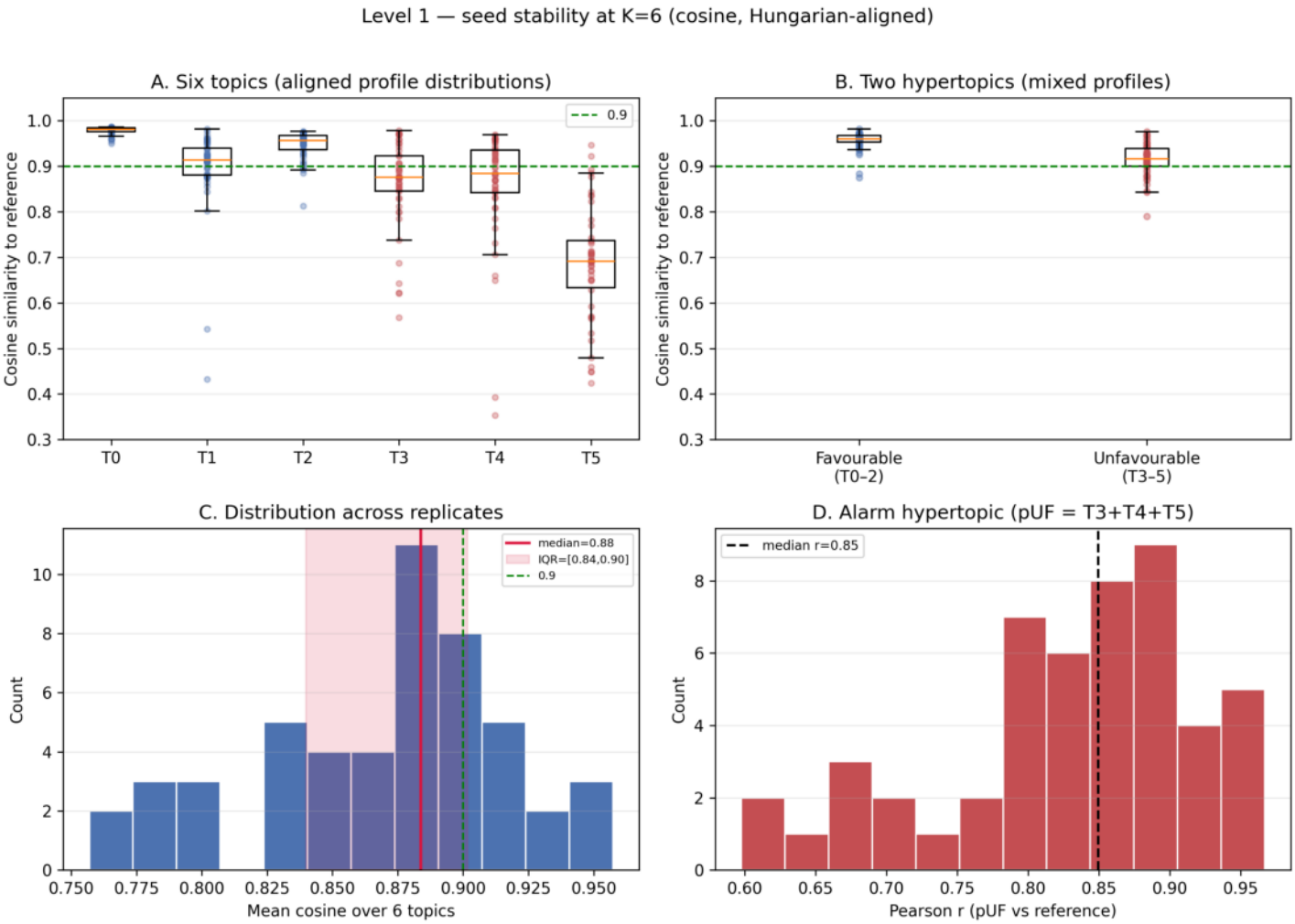
Headline (cosine)

Summary	Value
Median cosine (all topic–rep pairs)	0.915
IQR	[0.844, 0.962]
Median of per-replicate mean cosine	0.884
IQR of per-replicate mean cosine	[0.840, 0.902]
Favourable hypertopic cosine (median)	0.960
Unfavourable hypertopic cosine (median)	0.916
Patient pUF Pearson r (median [IQR])	0.849 [0.795, 0.890]

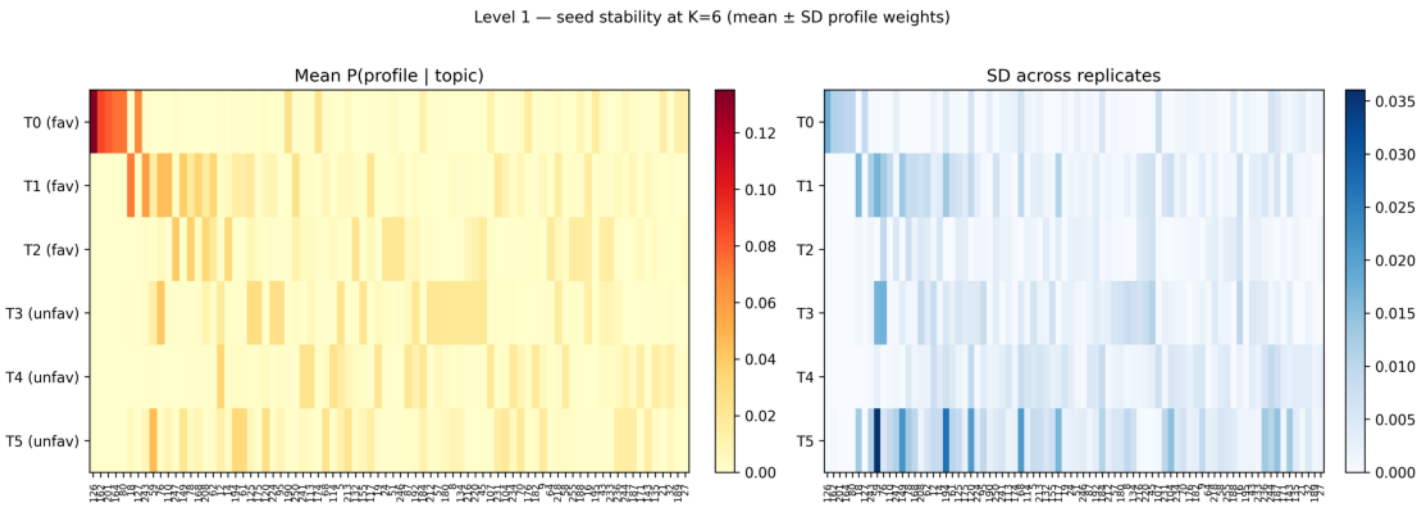
Per-topic cosine

Topic	Hypertopic	Cosine median [IQR]
T0	favourable	0.981 [0.976, 0.984]
T1	favourable	0.914 [0.880, 0.940]
T2	favourable	0.957 [0.936, 0.967]
T3	unfavourable	0.875 [0.846, 0.923]
T4	unfavourable	0.884 [0.843, 0.935]
T5	unfavourable	0.691 [0.633, 0.737]

Level 1 — cosine



Level 1 — profile heatmap



3. Level 2 — leave-20%-out stability

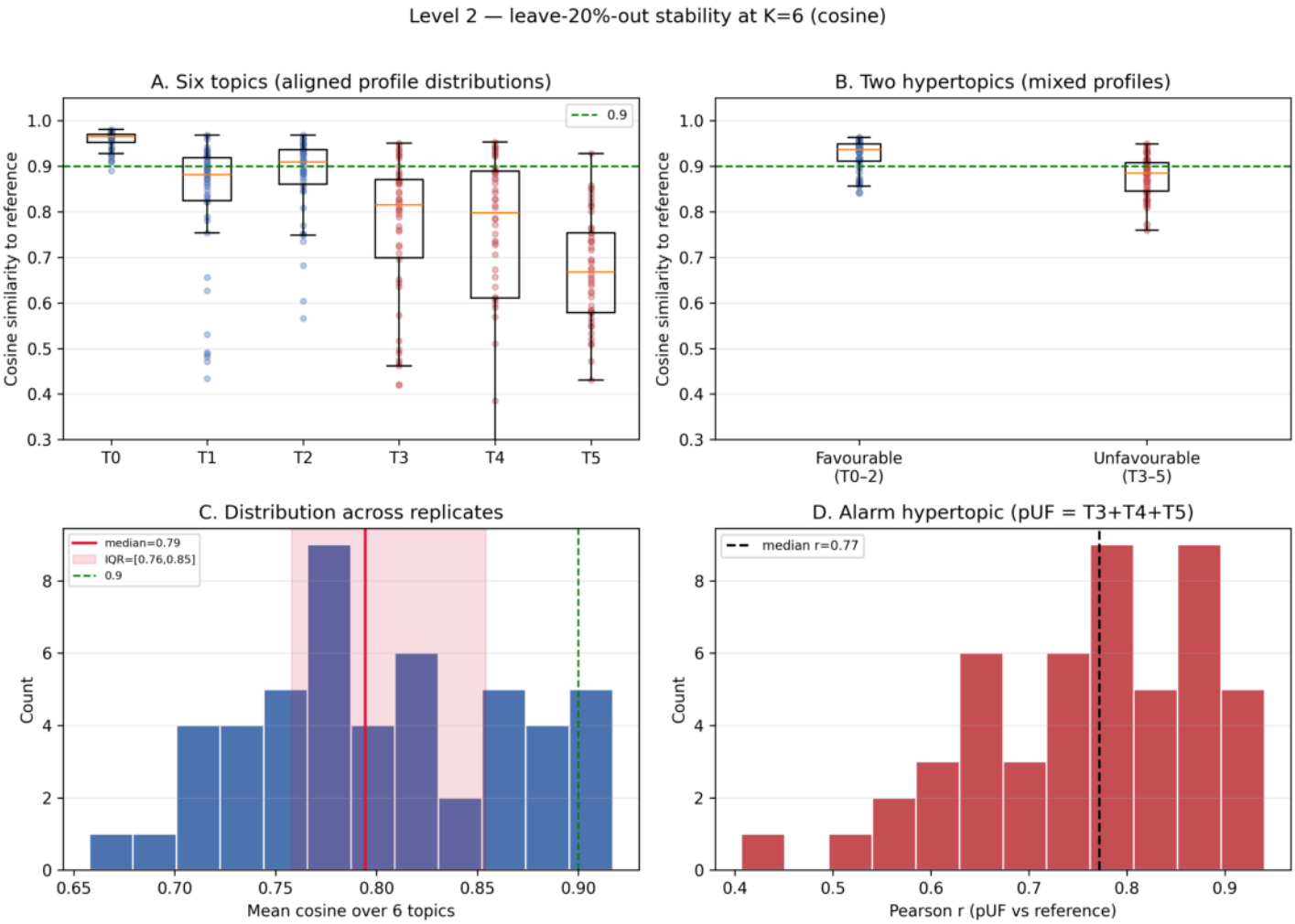
Headline (cosine)

Summary	Value
Median cosine (all topic–rep pairs)	0.867
IQR	[0.723, 0.931]
Median of per-replicate mean cosine	0.794
Fav / unfav hypertopic cosine	0.937 / 0.885
Patient pUF Pearson r (median [IQR])	0.772 [0.675, 0.858]

Per-topic cosine

Topic	Hypertopic	Cosine median [IQR]
T0	favourable	0.965 [0.953, 0.970]
T1	favourable	0.882 [0.825, 0.919]
T2	favourable	0.909 [0.860, 0.937]
T3	unfavourable	0.815 [0.699, 0.871]
T4	unfavourable	0.797 [0.611, 0.889]
T5	unfavourable	0.668 [0.579, 0.753]

Level 2 — cosine



Level 2 — profile heatmap

